1.7.8 Advanced Topic: Lists as Sets

Mathematica usually keeps the elements of a list in exactly the order you originally entered them. If you want to treat a *Mathematica* list like a mathematical *set*, however, you may want to ignore the order of elements in the list.

`Union[`*list1*`,` *list2*`, ...]`	give a list of the distinct elements in the $list_i$
`Intersection[`*list1*`,` *list2*`, ...]`	give a list of the elements that are common to all the $list_i$
`Complement[`*universal*`,` *list1*`, ...]`	give a list of the elements that are in *universal*, but not in any of the $list_i$

Set theoretical functions.

`Union` gives the elements that occur in *any* of the lists.

```
In[1]:= Union[{c, a, b}, {d, a, c}, {a, e}]
Out[1]= {a, b, c, d, e}
```

`Intersection` gives only elements that occur in *all* the lists.

```
In[2]:= Intersection[{a, c, b}, {b, a, d, a}]
Out[2]= {a, b}
```

`Complement` gives elements that occur in the first list, but not in any of the others.

```
In[3]:= Complement[{a, b, c, d}, {a, d}]
Out[3]= {b, c}
```

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